Secure Network Design

Matadors University

8 Colleges · 642 Faculty · ~2,568 Devices · Wired + Wi-Fi · $200K

IS457 Advanced Telecommunications and Networking

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Agenda — 20 minutes

Four parts, five minutes each

1

Problem & Design Overview

Scenario, requirements, key assumptions, and high-level architecture

Bounam Aphayavong

5 min

2

IP Addressing, VLANs & Wi-Fi

Subnetting, VLAN-to-college mapping, DHCP and wireless infrastructure

Arianna Morales

5 min

3

Routing, STP, EtherChannel & Security

Inter-VLAN routing, RSTP, link aggregation, ACLs, port security

Dylan Colazo Douriet

5 min

4

Budget, Testing & Conclusion

Hardware costs, validation tests, lessons learned, Q&A

Harrison Argueta

5 min

Part 1

The Problem & Our Design

Scenario · Requirements · Architecture · Assumptions

Bounam Aphayavong — 5 minutes

The Scenario

Eight colleges, one secure campus network

College	Faculty	Devices (×4)	Relative size
Humanities	119	**476**	
Business and Economics	102	**408**	
Social Sciences	92	**368**	
Education	80	**320**	
Science and Mathematics	70	**280**	
Engineering	69	**276**	
Art and Media	65	**260**	
Health and Human Sciences	45	**180**	

Total: 642 faculty · ~2,568 devices · *each faculty has 4 devices: PC + Laptop + IP Phone + Printer access*

Requirements & Constraints

What the brief asks us to deliver

Functional requirements

Network segmentation
One dedicated subnet and VLAN per college

Inter-VLAN routing
Controlled communication between colleges

Redundancy & loop prevention
STP root bridge + EtherChannel link aggregation

Security controls
ACLs, port security, device hardening

Project constraints

$200,000

Total budget cap (hardware + labor + training)

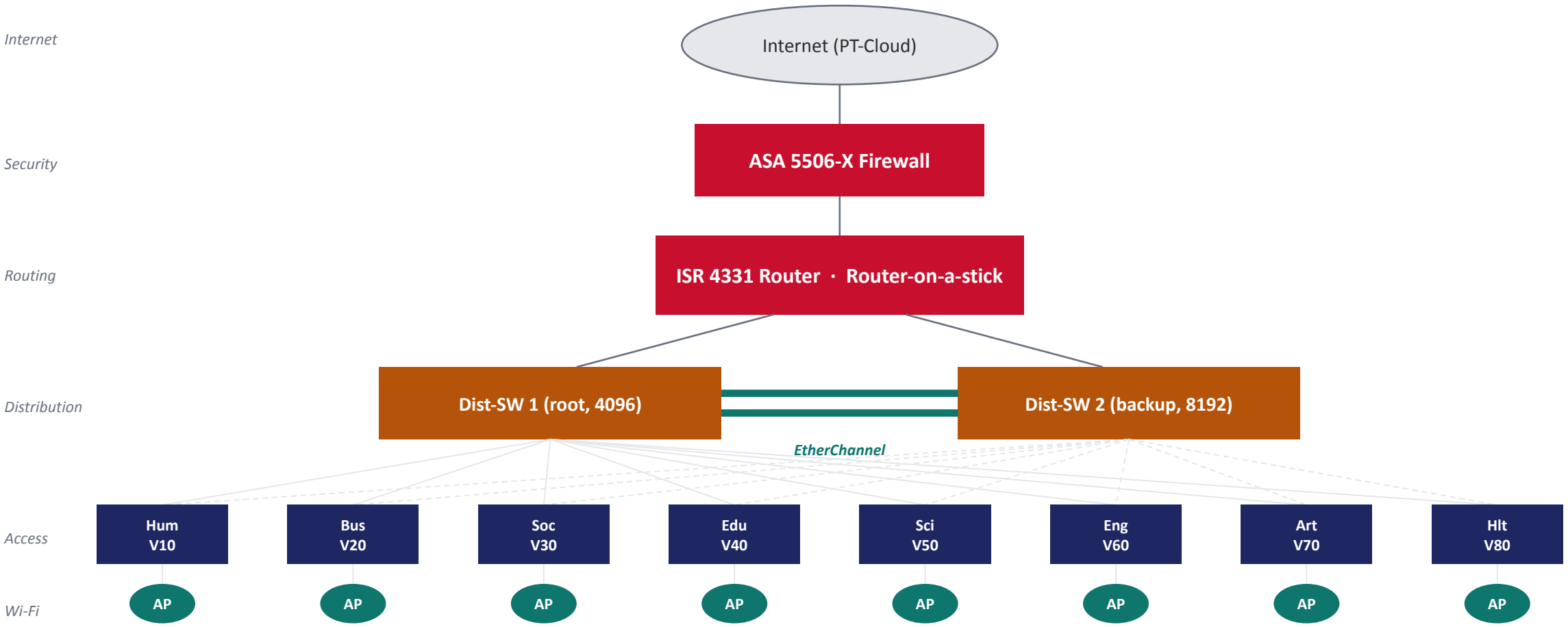
Static IP addressing
No DHCP — every device manually assigned

Cisco Packet Tracer simulation
Mandatory deliverable for the report

Vendor-cited pricing
All costs traceable to verifiable sources

High-Level Architecture

Three-tier hierarchical design with redundancy



Each access switch is dual-homed to both Dist switches — STP blocks one path until failover. Each college has its own Wi-Fi AP on its VLAN.

Key Assumptions

What we assumed to make the design feasible

1. **Four devices per faculty member**

   PC, laptop, IP phone, and shared printer access — totaling ~2,568 devices across the campus.

2. **Private addressing with 10.0.0.0/16**

   Sized for growth. Public IPs handled at the firewall via NAT — internal devices remain private.

3. **Multiple stacked switches per college**

   A 48-port switch holds 44 usable ports. Larger colleges need up to 11 switches daisy-chained.

4. **Single physical site**

   All eight colleges are on one campus. No WAN links between distant sites are required.

5. **Hybrid IP addressing: static + DHCP**

   Static IPs for fixed devices (PCs, IP phones, printers, APs); DHCP pools on the router for mobile Wi-Fi clients (laptops, smartphones, tablets).

Part 2

IP, VLANs, DHCP & Wi-Fi

Subnetting · VLAN mapping · DHCP pools · Wireless access

Arianna Morales — 5 minutes

Subnet Sizing Logic

Why /23 for most colleges and /24 for the smallest

/23

510 usable hosts
Mask: 255.255.254.0

Used for 7 of 8 colleges

/24

254 usable hosts
Mask: 255.255.255.0

Used for Health & Human Sci. (180 dev)

The reasoning

Largest college (Humanities) needs 476 devices.
→ /24 (254) is too small. /23 (510) gives ~7% growth headroom.
Smallest college (Health) needs 180 devices.
→ /24 (254) gives 41% headroom — efficient use of address space.
Each subnet split: x.x.x.1-99 static (PCs, phones, printers, APs) + x.x.x.100-x.x.x.253 DHCP (Wi-Fi).

VLAN-to-Subnet Mapping

One VLAN per college · gateway = last usable IP

VLAN	College	Subnet	Usable Range	Gateway	Devices
10	Humanities	10.0.0.0/23	10.0.0.1 → 10.0.1.254	10.0.1.254	476
20	Business & Economics	10.0.2.0/23	10.0.2.1 → 10.0.3.254	10.0.3.254	408
30	Social Sciences	10.0.4.0/23	10.0.4.1 → 10.0.5.254	10.0.5.254	368
40	Education	10.0.6.0/23	10.0.6.1 → 10.0.7.254	10.0.7.254	320
50	Science & Math	10.0.8.0/23	10.0.8.1 → 10.0.9.254	10.0.9.254	280
60	Engineering	10.0.10.0/23	10.0.10.1 → 10.0.11.254	10.0.11.254	276
70	Art & Media	10.0.12.0/23	10.0.12.1 → 10.0.13.254	10.0.13.254	260
80	Health & Human Sci.	10.0.14.0/24	10.0.14.1 → 10.0.14.254	10.0.14.254	180

Why these specific subnets? Sequential allocation aligned to /23 boundaries (3rd octet must be even). *No gaps, no overlaps, room for future VLANs.*

How a VLAN Works

Logical separation of devices on the same physical infrastructure

Step 1

Access port

Each port on an access switch is assigned to one VLAN (e.g. VLAN 10).
Devices plugged into that port belong to VLAN 10 — they only see other VLAN 10 devices.

Step 2

Trunk port (802.1Q tagging)

Inter-switch links carry traffic for ALL VLANs.
Each Ethernet frame gets a 4-byte 802.1Q tag identifying its VLAN, so switches know how to deliver it.

Step 3

Inter-VLAN routing

VLANs cannot talk to each other directly.
Traffic must pass through a router (or L3 switch) where ACLs decide what's allowed.

VLAN Configuration — CLI Example

Same commands repeated on every access switch with the appropriate VLAN ID

Create VLANs (on every switch)

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Humanities
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Business
...
Switch(config-vlan)# vlan 80
Switch(config-vlan)# name Health
Switch(config-vlan)# exit
Switch# show vlan brief
```

Assign access ports + trunk uplink

```
! On SW-Humanities – assign ports to VLAN 10
Switch(config)# interface range fa0/1-44
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# spanning-tree portfast
Switch(config-if)# switchport port-security

! Configure trunk uplink to Dist-Switch
Switch(config)# interface gi0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed
                   vlan 10,20,30,40,50,60,70,80
```

Wireless Infrastructure

One Wi-Fi access point per college, each on the local VLAN

How it works

Each college has 1 dedicated AP:
AP-Humanities, AP-Business, etc.

Each AP plugs into Fa0/4 of its college switch
The switch port is in access mode VLAN X

Each AP broadcasts a unique SSID
Matadors-Humanities, Matadors-Business, ...

WPA2-PSK with AES encryption
Same security policies as wired devices

DHCP for mobile clients

DHCP server hosted on the router
8 pools, one per VLAN, on the ISR 4331

Static range: x.x.x.1 to x.x.x.99
PCs, IP phones, printers, APs

DHCP range: x.x.x.100 to x.x.+1.253
Laptops, smartphones, tablets

Lease: 7 days · DNS: 8.8.8.8

```
Router(config)# ip dhcp pool POOL_HUMANITIES
```

Part 3

Routing, STP & Security

Inter-VLAN routing · RSTP · EtherChannel · ACLs · Port security

Dylan Colazo Douriet — 5 minutes

Inter-VLAN Routing — Router on a Stick

One physical interface, eight logical sub-interfaces, eight VLAN gateways

The concept

ISR 4331 Router · Gi0/0/1

```
Gi0/0/1.10  →  10.0.1.254 (VLAN 10)
Gi0/0/1.20  →  10.0.3.254 (VLAN 20)
Gi0/0/1.30  →  10.0.5.254 (VLAN 30)
Gi0/0/1.40  →  10.0.7.254 (VLAN 40)
Gi0/0/1.50  →  10.0.9.254 (VLAN 50)
Gi0/0/1.60  →  10.0.11.254 (VLAN 60)
Gi0/0/1.70  →  10.0.13.254 (VLAN 70)
Gi0/0/1.80  →  10.0.14.254 (VLAN 80)
```

Sample sub-interface configuration

```
Router(config)# interface gi0/0/1
Router(config-if)# no shutdown

Router(config)# interface gi0/0/1.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 10.0.1.254
                      255.255.254.0

Router(config)# interface gi0/0/1.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 10.0.3.254
                      255.255.254.0

! Repeat for VLANs 30 through 80
```

Loop Prevention & Link Aggregation

RSTP eliminates loops · EtherChannel doubles bandwidth on critical links

RSTP — Rapid Spanning Tree

Why? Redundant links create broadcast loops that collapse the network.

How? RSTP elects a root bridge and blocks redundant paths until needed.

Our config:
- Dist-SW 1 = root (priority 4096)
- Dist-SW 2 = backup (priority 8192)
- Per-VLAN STP instances (rapid-pvst)
- PortFast on all access ports

```
Switch(config)# spanning-tree mode rapid-pvst
```

EtherChannel — Link Aggregation

Why? STP would block one of two parallel links — wasted bandwidth.

How? Bundle multiple physical links into one logical link STP sees as a single connection.

Our config:
- Between Dist-SW 1 and Dist-SW 2
- 2 Gigabit links ⟶ 2 Gbps logical link
- Static mode (channel-group 1 mode on)
- Survives single-link failure transparently

```
Switch(config-if)# channel-group 1 mode on
```

Security in Depth

Four layers of defense across the network

1

Perimeter — ASA 5506-X firewall

NAT, stateful inspection, and URL filtering between campus and Internet.

2

Network — Inter-VLAN ACLs on the router

Permit/deny rules between colleges (e.g. block direct Humanities ↔ Engineering traffic).

3

Switch — Port security on access ports

Maximum 2 MAC addresses per port. Violation triggers shutdown to prevent rogue devices.

4

Device — Hardening on every Cisco device

SSH only (no Telnet), strong enable secret, disabled unused ports, banner with legal notice.

Sample ACL — Inter-VLAN Restriction

Block direct traffic between Humanities and Engineering, while allowing both to reach shared services

Extended ACL on the router

```
Router(config)# ip access-list extended BLOCK_HUM_ENG
Router(config-ext-nacl)# deny ip 10.0.0.0 0.0.1.255
                                 10.0.10.0 0.0.1.255
Router(config-ext-nacl)# deny ip 10.0.10.0 0.0.1.255
                                 10.0.0.0 0.0.1.255
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit

Router(config)# interface gi0/0/1.10
Router(config-subif)# ip access-group BLOCK_HUM_ENG in
Router(config)# interface gi0/0/1.60
Router(config-subif)# ip access-group BLOCK_HUM_ENG in
```

How it reads

Line 1-2:
Block Humanities ⟶ Engineering traffic

Line 3-4:
Block Engineering ⟶ Humanities traffic

Line 5:
Allow everything else (other VLANs, Internet)

Wildcard mask 0.0.1.255 = /23

Result in Packet Tracer: Ping from Humanities PC (10.0.0.5) to Engineering PC (10.0.10.5) ⟶ Request timed out. *Ping to Education PC (10.0.6.5) ⟶ succeeds.*

Part 4

Budget, Testing & Conclusion

Hardware costs · Validation · Lessons learned

Harrison Argueta — 5 minutes

Budget Breakdown

Vendor-sourced prices from Amazon, ITPrice, and Best Buy (May 2026)

Category	Item	Qty	Unit ($)	Subtotal ($)
Network Core	1× Cisco ISR 4331 Router	1	$895	**$895**
	1× Cisco ASA 5506-X Firewall	1	$549	**$549**
	2× Cisco 2960-48TT-L (Distribution)	2	$219	**$438**
Access Layer	63× Cisco 2960-48TT-L (Access)	63	$219	**$13,797**
Wireless	25× Cisco AP 3702i (lightweight)	25	$99	**$2,475**
IP Telephony	642× Cisco IP Phones (7821 series)	642	$117	**$75,114**
Printing	32× HP LaserJet 3301fdw printers	32	$639	**$20,448**
Cabling	Cat 6 (14× 1000ft) + racks + patch panels	1	$4,900	**$4,900**
Services	Installation labor & configuration	1	$16,000	**$16,000**
	First-year maintenance + training	1	$8,000	**$8,000**
PROJECT TOTAL (within $200,000 budget)				**$142,616**

Desktops and laptops procured separately by each college's IT budget. Switches and IP phones sourced as Cisco-certified refurbished, a standard practice in academic procurement to reduce capital expenditure while maintaining vendor support.

Testing & Validation

How we proved the design works in Cisco Packet Tracer

Intra-VLAN ping (wired)

`PC-Hum → PC-Hum (same VLAN)`

✓ **Reply received**

Inter-VLAN ping (allowed)

`ping 10.0.6.5 (Hum → Edu)`

✓ **Reply via router**

Inter-VLAN ping (blocked)

`ping 10.0.10.5 (Hum → Eng)`

✗ **Request timed out (ACL)**

DHCP attribution (Wi-Fi)

`ipconfig on Wi-Fi laptop`

✓ **IP from x.x.x.100+ pool**

Wi-Fi intra-VLAN ping

`Laptop Wi-Fi → PC wired`

✓ **Same VLAN, reply OK**

Wi-Fi blocked by ACL

`Hum laptop → Eng PC`

✗ **ACL blocks Wi-Fi too**

STP root verification

`show spanning-tree vlan 10`

✓ **Root = Dist-SW 1 (4096)**

EtherChannel + Failover

`Cable test + show etherchannel`

✓ **Po1/SU, <1s reconverge**

All tests passed — wired and wireless behave identically.

Connectivity, segmentation, redundancy, DHCP, and security policies all enforced across both connection types.

Lessons Learned & Future Work

What we'd do differently and where this design could grow

Lessons Learned

Subnet alignment matters.
/23 subnets must start on even third-octet values.

VLANs ≠ subnets, but should align 1-to-1.
One VLAN per subnet keeps the design clean.

STP is invisible until it isn't.
Set explicit root priority — don't rely on MAC-based election.

ACLs are powerful but unforgiving.
One wrong wildcard mask blocks more than intended.

Future Enhancements

WPA2-Enterprise with 802.1X
Replace shared Wi-Fi password with per-user auth.

Wireless LAN Controller
Centralized management for all 25 access points.

Stackable Catalyst 2960-X switches
Unified management when budget allows.

VPN access for remote faculty
IPsec or SSL VPN through the ASA firewall.

Network monitoring (SNMP, syslog, NetFlow)

Questions?

Thank you for your attention.

Bounam Aphayavong · Arianna Morales · Dylan Colazo Douriet · Harrison Argueta

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